

p-Adic Metrics and the Infinitude of Primes

MATH200

Doruk Üstündağ

Izmir Institute of Technology

09.04.2025



Outline

- 1 A Brief History
- 2 Introduction
- 3 p -adic Metric
- 4 Geometric Proof of Infinitude of Primes
- 5 References

Historical Approach

- Euclid
- Euler
- Erdős
- Furstenberg
- First geometric proof by Haydar Göral
- ...

Generality of Notations

- p always denotes a prime number.
- Set of all primes are denoted by $\mathbb{P}$.

Definition

p-adic order (valuation) of a given non-zero integer n denoted by $\nu_p(n)$ is the largest power of p dividing n . In other words, for a non-zero integer n , the equality $\nu_p(n) = m$ means that p^m divides n but p^{m+1} does not divide n .

$$\nu_p(n) = \begin{cases} \max\{k \in \mathbb{N}_0 : p^k \mid n\} & \text{if } n \neq 0 \\ \infty & \text{if } n = 0, \end{cases}$$

Example: p-adic Order

Example

$$\nu_2(2 \cdot 3^2) = \nu_2(18) = \nu_2(-18) = 1$$

$$\nu_3(2 \cdot 3^2) = \nu_3(18) = \nu_3(-18) = 2$$

Definition

p-adic absolute value of a non-zero integer n is defined as:

$$|n|_p = p^{-\nu_p(n)},$$

and $|0|_p = 0$ as usual.

Example: p-adic Absolute Value

Example

$$|18|_2 = |-18|_2 = 2^{-\nu_2(18)} = 2^{-1} = \frac{1}{2}$$

$$|18|_3 = |-18|_3 = 3^{-\nu_3(18)} = 3^{-2} = \frac{1}{9}$$

More on p-adic Absolute Value

To clear few things out we can see these as observations:

$$|1|_p = 1 \text{ and } |p|_p = \frac{1}{p}$$

for any $p \in \mathbb{P}$. If $p|a$, then $|a|_p \leq \frac{1}{p}$, and if $p \nmid a$, then $|a|_p = 1$.
Trivially we can see that:

$$0 \leq |a|_p \leq 1 \text{ and } |a|_p = |-a|_p$$

for any integer a and prime p .

Definition (Metric)

Let X be a non-empty set. A **metric** d on X is a map $d : X \times X \rightarrow [0, \infty)$ such that the following three conditions are satisfied:

- (Positivity) For all $x, y \in X$, $d(x, y) \geq 0$ and $d(x, y) = 0 \iff x = y$.
- (Symmetry) For all $x, y \in X$, $d(x, y) = d(y, x)$.
- (Triangle Inequality) For all $x, y, z \in X$, $d(x, y) \leq d(x, z) + d(z, y)$.

Definition (Metric Space)

If d is a metric on X , then the pair (X, d) is called a **metric space**.

Definition

The **p-adic metric** between integers x and y is defined as:

$$d_p(x, y) = |x - y|_p$$

Proposition 1

For any prime number p , the map d_p is a metric on $\mathbb{Z}$.

To prove that d_p is a metric on $\mathbb{Z}$ we should show that it satisfies three properties of being a metric space.

- Positivity
- Symmetry
- Triangle Inequality

Positivity

Let x and y be two integers. From definitions we can see that,

$$d_p(x, y) = |x - y|_p \geq 0 \text{ and } d_p(x, x) = |x - x|_p = |0|_p = 0.$$

Suppose $d_p(x, y) = 0$. We must show that $x = y$. If x is different from y , then $x - y \neq 0$, and there exists a non-negative integer m such that p^m divides $x - y$ but p^{m+1} does not divide $x - y$. We have $\nu_p(x - y) = m$ and so

$$d_p(x, y) = |x - y|_p = p^{-m} \neq 0$$

which is a contradiction to our assumption. We conclude that

$$d_p(x, y) = 0 \implies x = y$$

Symmetry

Let x and y be two integers. If $x = y$, then it is trivial:

$$d_p(x, y) = 0 = d_p(y, x)$$

If $x \neq y$, we can see that by the properties of p-adic order

$$\nu_p(x - y) = \nu_p(y - x),$$

and it follows again that $d_p(x, y) = p^{-\nu_p(x-y)} = p^{-\nu_p(y-x)} = d_p(y, x)$.

Triangle Inequality

Let x, y and z be three integers. If $x = y, x = z$, or $y = z$, then the triangle inequality follows either trivially or from positivity. We assume that x, y and z are distinct integers. Let

$$m = \nu_p(x - y), n = \nu_p(x - z), \text{ and } k = \nu_p(z - y).$$

Suppose that $n \leq k$. Since

$$x - y = (x - z) + (z - y)$$

and p^n divides $x - z$ and $z - y$, we see that p^n also divides $(x - y)$. This shows that $n \leq m$. Thus,

$$d_p(x, y) = p^{-m} \leq p^{-n} \leq p^{-n} + p^{-k} = d_p(x, z) + d_p(z, y).$$

Proposition 2

Let X be a non-empty set and $\rho_1, \dots, \rho_k$ be metrics on X . Then the map

$$\rho(x, y) = \rho_1(x, y) + \dots + \rho_k(x, y)$$

is also a metric on X

Example

$$\begin{aligned} d(x, y) &= d_2(x, y) + d_3(x, y) + d_5(x, y) + d_7(x, y) \\ &= |x - y|_2 + |x - y|_3 + |x - y|_5 + |x - y|_7 \end{aligned}$$

is a metric on $\mathbb{Z}$

Geometric Proof of Infinitude of Primes

Theorem

There are infinitely many primes.

Proof

Suppose that $\mathbb{P}$ is finite. So we can list all primes as $p_1, \dots, p_k$ where $k \geq 1$. For $a, b \in \mathbb{Z}$, define:

$$d(a, b) = \sum_{p \in \mathbb{P}} d_p(a, b) = \sum_{p \in \mathbb{P}} |a - b|_p = \sum_{i=1}^k |a - b|_{p_i}$$

Note that the sum above is actually finite as implied by the last equality.

Geometric Proof of Infinitude of Primes

Proof (Cont.)

By Propositions 1 & 2, we can see that d is a metric on $\mathbb{Z}$. For any integer m , we have

$$d(m, m+1) = d(m, m-1) = \sum_{p \in \mathbb{P}} |1|_p = \sum_{p \in \mathbb{P}} 1 = k$$

If $n \notin \{m-1, m, m+1\}$, then the integer $m-n$ is of absolute value greater than or equal to 2, and hence it is divisible by some prime q .

$$|m-n|_q \leq \frac{1}{q} \leq \frac{1}{2}$$

Geometric Proof of Infinitude of Primes

Proof (Cont.)

Therefore, if $n \notin \{m-1, m, m+1\}$ then

$$\begin{aligned} d(m, n) &= |m - n|_q + \sum_{p \in \mathbb{P} - \{q\}} |m - n|_p \\ &\leq \frac{1}{q} + \sum_{p \in \mathbb{P} - \{q\}} 1 \leq \frac{1}{2} + k - 1 = k - \frac{1}{2}. \end{aligned}$$

Proof (Cont.)

Moreover,

$$\begin{aligned}d(m-1, m+1) &= \sum_{p \in \mathbb{P}} |2|_p = |2|_2 + \sum_{p \in \mathbb{P} - \{2\}} |2|_p \\ &= \frac{1}{2} + k - 1 = k - \frac{1}{2}\end{aligned}$$

Geometric Proof of Infinitude of Primes

Proof (Cont.)

We define the open ball of radius $r > 0$ and center $x \in \mathbb{Z}$ to be the set

$$B(x, r) = \{y \in \mathbb{Z} : d(x, y) < r\}.$$

Based on the observations, we now claim that for any integer m , we have

$$B\left(m + 1, \frac{1}{2}\right) = \{m + 1\}.$$

To see this, notice that we clearly have $m + 1$ is in $B(m + 1, \frac{1}{2})$. Let $n \neq m + 1$ be in $B(m + 1, \frac{1}{2})$, which is to say that $d(m + 1, n) < \frac{1}{2}$.

Geometric Proof of Infinitude of Primes

Proof (Cont.)

Note that the above observations imply that n cannot be m or $m - 1$ since $k > k - \frac{1}{2} \geq \frac{1}{2}$. Thus, we may assume that n is different from $m - 1, m, m + 1$. Then by the equation, we have that $d(m, n) \leq k - \frac{1}{2}$. Applying triangle inequality to d , we get that

$$k = d(m, m + 1) \leq d(m, n) + d(n, m + 1) < k - \frac{1}{2} + \frac{1}{2} = k,$$

a contradiction. This shows that claim is true.

Geometric Proof of Infinitude of Primes

Proof (Cont.)

However, we can also show that the claim is not true. Choose positive integers b_i such that

$$\frac{1}{p_i^{b_i}} < \frac{1}{2k}$$

for $i = 1, \dots, k$. Set $\ell = p_1^{b_1} \dots p_k^{b_k}$. Then

$$d(\ell, 0) = \frac{1}{p_1^{b_1}} + \dots + \frac{1}{p_k^{b_k}} < k \cdot \frac{1}{2k} = \frac{1}{2}.$$

This contradicts the claim above for $m = \ell - 1$, and hence establishes that there are infinitely many primes. □

- Göral, Haydar. (2020). p -Adic Metrics and the Infinitude of Primes. Mathematics Magazine. 93. 19-22.
10.1080/0025570X.2020.1679574.